

Water Electrolysis

Experiment Packet

Audience: instructors and teaching assistants

Purpose: Bench instructions for instructors and TAs

Session hook: Split water and see a 2:1 ratio

Length: 90-minute lab block (~20 min concepts + ~70 min hands-on)

This packet includes

1. Session overview & plan
2. Preparation
3. Materials checklist
4. Experiment procedure
5. Shared safety notes

Generated from the Electrochemistry workshop curriculum. Prefer the latest PDF from the course repository if procedures change mid-week.

Session 4 — Water Electrolysis and Gas Collection

Hook: Split water, collect gases, and see a 2:1 stoichiometric ratio.

Learning outcomes

- Run water electrolysis in a Hoffman-style **U-tube** with graphite electrodes
- Use **sodium sulfate** electrolyte safely (never NaCl)
- Measure H₂ and O₂ volumes and compare to 2:1 ratio
- Connect electrolysis to hydrogen as an energy storage concept
- **Start** electrolytic rust-removal cells for Session 5 reveal

Session plan (90 min)

Time	Activity	Reference
0–10 min	Hook: U-tube cell — which limb fills with gas twice as fast?	lecture.md
10–25 min	Water splitting and stoichiometry	lecture.md
25–40 min	Fill and wire the Na ₂ SO ₄ U-tube apparatus	experiment.md — Part A
40–60 min	Run electrolysis; record volumes over time	experiment.md — Part B
60–70 min	Discuss ratio; optional instructor H ₂ pop test	experiment.md — Part C
70–80 min	Brief energy storage discussion	lecture.md
80–90 min	Start derusting cells for Session 5	experiment.md — Part E

Key reactions

Overall: $2\text{H}_2\text{O}(\text{l}) \rightarrow 2\text{H}_2(\text{g}) + \text{O}_2(\text{g})$

Cathode (–): $2\text{H}_2\text{O} + 2\text{e}^- \rightarrow \text{H}_2 + 2\text{OH}^-$

Anode (+): $2\text{H}_2\text{O} \rightarrow \text{O}_2 + 4\text{H}^+ + 4\text{e}^-$ (acidic media)

or $4\text{OH}^- \rightarrow \text{O}_2 + 2\text{H}_2\text{O} + 4\text{e}^-$ (basic media)

Status

- [] U-tube apparatus + graphite electrodes sourced
- [] ~0.1 M Na₂SO₄ mixed and labeled
- [] Apparatus pilot-tested (2:1 visible)
- [] **No saltwater** — Na₂SO₄ only for electrolysis
- [] Pop test protocol reviewed
- [] Rusty nails + washing soda ready for Part E

Session 4 — Preparation

Before class (instructor)

- [] **Pilot the U-tube apparatus** — confirm 2:1 ratio visible within session time
- [] Check graphite electrodes, stoppers, and spade connectors
- [] Mix **~0.1 M Na₂SO₄** batch; label **"NO SALT"**
- [] Review pop test safety; decide if demo will run (weather, room rules)
- [] Optional: test solar panel output with the same cell
- [] Stage rusty nails, washing soda, sacrificial anodes, and spare jars for **Part E** (end of class)
- [] Confirm overnight power policy for derusting cells (prep room vs. disconnect)

Day-of setup

- [] One U-tube station per group (or shared stations if kits are limited)
- [] Pre-mix Na₂SO₄; aliquot or fill stations
- [] Large sign: **"No salt in electrolysis — use sodium sulfate"**
- [] Separate instructor area for pop test marked off
- [] Part E station: rusty nails, washing soda, steel/graphite anodes, labels, camera

Common failure modes (from pilot)

Issue	Prevention
Weak bubble rate	Fresh ~0.1 M Na ₂ SO ₄ ; clean graphite; firm spade connections
Students add salt "to speed up"	Pre-brief; labeled Na ₂ SO ₄ only
Overpressure	Keep side arms open or vented
Reversed polarity	Mark (–)=H ₂ / (+)=O ₂ on the stand before power-on

Open questions / notes

- Supply voltage used:
- Best volume-measurement method (graduations vs. ruler):
- Pop test yes/no for this venue:

Session 4 — Materials

Per group / station

- ☐ Hoffman-style glass **U-tube** with side arms + acrylic stand
- ☐ Two graphite electrodes with rubber/cork stoppers
- ☐ Red (+) / black (–) leads with spade connectors
- ☐ DC power supply or battery pack (6–12 V typical)
- ☐ **Sodium sulfate** electrolyte: **~0.1 M Na₂SO₄** (~7 g anhydrous / 500 mL; ~3.5 g / 250 mL)
- ☐ Tap water for mixing
- ☐ Ruler (gas column / liquid-level comparison)
- ☐ Goggles

Optional

- ☐ pH / universal indicator
- ☐ Small solar panel
- ☐ Tubing for side-arm gas collection
- ☐ Baking soda (emergency backup electrolyte only)

Instructor-only (pop test)

- ☐ Small flame source (long lighter)
- ☐ Separate tiny collection vessel
- ☐ Fire extinguisher accessible

Part E — Start rust removal (end of class)

- ☐ Rusty iron nails or hardware (2+ per group)
- ☐ Washing soda: **1 tbsp per ~250 mL** warm water (preferred) **or** baking soda same measure (NOT NaCl)
- ☐ Sacrificial steel strips or graphite rods
- ☐ Small jars or cups
- ☐ Labels and camera for before photos

Explicitly excluded

- ☐ ~~Table salt (NaCl)~~ — **do not use** in electrolysis or derusting baths

Pilot notes

Parameter	Value from pilot
Electrolyte	~0.1 M Na ₂ SO ₄

Parameter	Value from pilot
Supply voltage	__ V
Time to visible gas difference	__ min
Typical V_{H_2}/V_{O_2}	__

Session 4 — Experiment: Water Electrolysis

Instructors: gas ID, ratio systematics, Part E polarity — instructor.md.

Electrolyte — sodium sulfate (Na_2SO_4)

Never use table salt (NaCl) — chloride can produce toxic chlorine gas at the anode.

Recipe — ~0.1 M Na_2SO_4

Batch	Na_2SO_4 (anhydrous)	Water
250 mL (about one U-tube fill)	~3.5 g	Fill to 250 mL
500 mL	~7 g	Fill to 500 mL
1.0 L	~14 g	Fill to 1.0 L

Stir until dissolved. Label "**0.1 M Na_2SO_4 electrolyte — NO SALT**".

Water: tap water is fine — Na_2SO_4 provides the ions you need. Distilled water is unnecessary here.

If you only have baking soda as emergency backup: 1–2 tsp NaHCO_3 per 500 mL works, but the planned electrolyte for this workshop is **sodium sulfate**.

Volume per station

Fill the U-tube so both arms are mostly full of electrolyte, with liquid well above the lower ends of the graphite electrodes and below the side-arm ports (see Part A). Mix **fresh** electrolyte if someone contaminated a batch with salt.

Part A — Assemble the U-tube electrolysis cell (25–40 min)

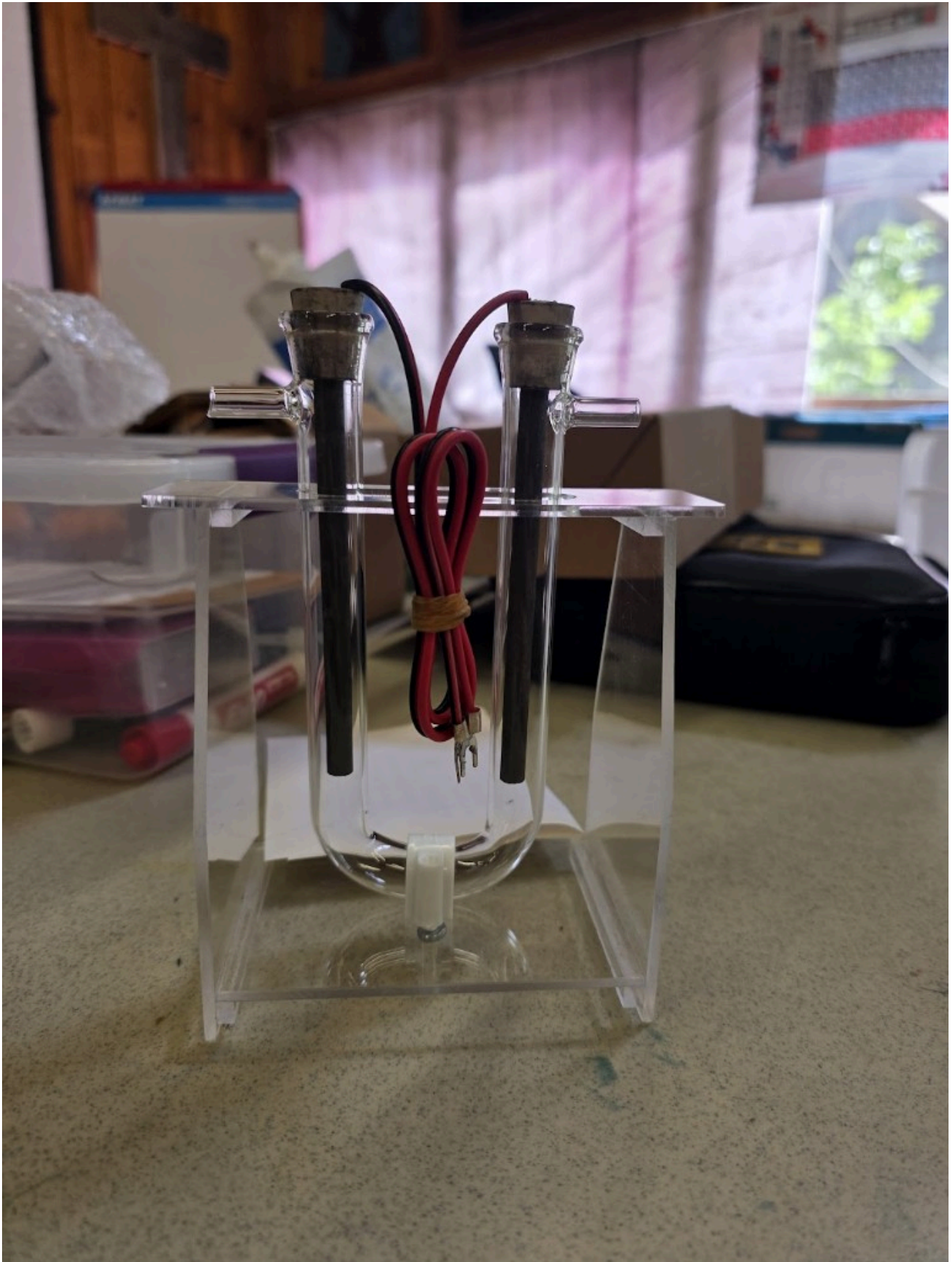


Figure 1 — Classroom apparatus: glass U-tube on an acrylic stand, graphite electrodes sealed with rubber stoppers, side arms on each limb, red (+) / black (–) leads with spade connectors for a DC supply.

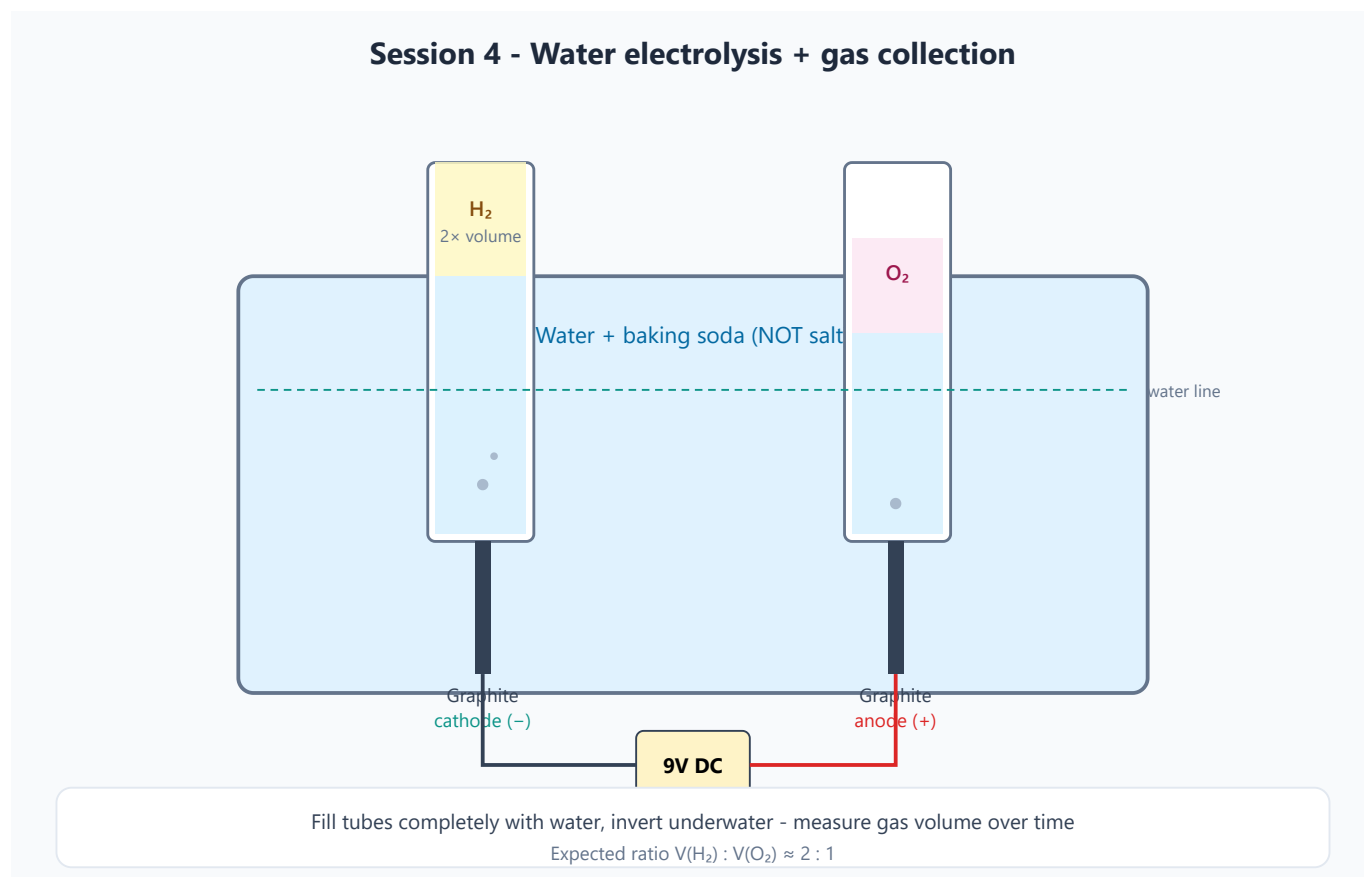


Figure 2 — Concept: H_2 forms at the cathode (–) at about twice the volume of O_2 at the anode (+). Electrolyte is sodium sulfate — not salt.

Materials

- Hoffman-style **glass U-tube** with **side arms** (one per limb), on acrylic stand
- Two **graphite** electrodes with rubber/cork stoppers
- Red (+) and black (–) leads with spade (fork) connectors
- DC power supply or battery pack (typically **6–12 V**; 9 V packs also work if current is adequate)
- **0.1 M Na_2SO_4** electrolyte
- Ruler or markings to compare gas column heights / liquid levels
- Goggles

Build steps

1. Place the U-tube securely in its acrylic stand.
2. Fill with **Na_2SO_4** electrolyte until both arms are nearly full — liquid should cover the immersed electrode tips and sit **below** the side-arm openings so gas can collect (or exit) there.
3. Insert one graphite electrode through each rubber stopper into each arm. Stoppers should seal the top of each limb.
4. Confirm electrodes do **not** touch each other or short through loose wires.

5. Connect leads:
6. **Black / (-)** → cathode → electrode that will collect **H₂** (faster-filling limb)
7. **Red / (+)** → anode → other electrode (**O₂**)
8. Attach spade connectors to the DC supply (or battery pack). Keep connections **dry**.
9. Side arms: leave open to air for classroom volume comparison, **or** attach collection tubing if your kit includes it. Do not seal both the stopper tops and the side arms airtight with no outlet — pressure must have a path.
10. When current flows, the **faster-rising gas column** (or faster liquid drop) is hydrogen (cathode).

Pre-run checklist

- [] U-tube filled with **Na₂SO₄** — **not salt**
- [] Stoppers seated; electrodes immersed and not touching
- [] Polarity marked: (-) = H₂ limb, (+) = O₂ limb
- [] Side arms open or properly tubed (no overpressure trap)
- [] Goggles on
- [] Power supply set; leads dry

Part B — Run and measure (40–65 min)

1. Start the timer when bubbles are clearly visible at both electrodes.
2. Every **2–3 min**, measure gas produced in each arm:
3. Prefer graduated limbs (mL) if marked, **or**
4. Measure **gas column height** or **liquid level drop** (cm) with a ruler (ratio ≈ volume ratio if the two arms match).
5. Continue until a clear difference appears or ~15–20 min have elapsed.

Gas volume log

Time (min)	V_H ₂ (mL or cm)	V_O ₂ (mL or cm)	Ratio V_H ₂ /V_O ₂
0	0	0	—
3			
6			
9			
12			
15			
20			

Expected result

V_H₂ / V_O₂ ≈ **2** (ideal). Classroom range **1.6–2.2** is a success — discuss solubility and collection errors.

Observations

- Bubble rate at each electrode
- Cloudiness or electrode crumbling
- Temperature change (optional)

Part C — Instructor-only pop test (65–75 min)

Optional demo — not for students

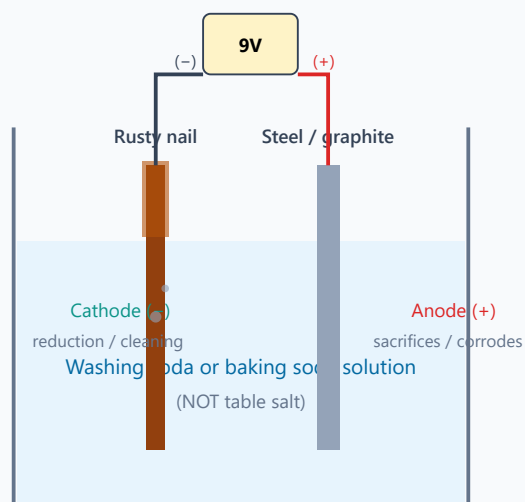
1. Collect a **tiny** amount of H_2 from the cathode side arm (or a small separate tube).
2. Move away from the main apparatus and faces.
3. Brief flame → soft “pop.”
4. Never ignite sealed or mixed H_2/O_2 containers.
5. Skip if the venue forbids open flame.

Part D — Extensions (if time)

- [] Add universal indicator — look for acid/base zones near electrodes
- [] Power with a small solar panel instead of a battery pack
- [] Swap polarity briefly: volumes reverse which limb grows faster (then restore)

Part E — Start electrolytic rust removal (80–90 min)

Electrolytic rust removal (start Session 4, reveal Session 5)



Start at end of Session 4; rinse and compare to control nail at start of Session 5
Same electrolysis idea as Session 4 - electricity drives a non-spontaneous reaction

Figure 3 — Start cells now; reveal cleaned nails at opening of Session 5.

Derusting electrolyte

Ingredient	Amount
Washing soda (sodium carbonate, Na_2CO_3) preferred	1 tbsp (~15 g) per ~250 mL warm water
or baking soda	1 tbsp per ~250 mL warm water

≈ **0.5–0.6 M** Na_2CO_3 if using washing soda at that kitchen measure — exact molarity is not critical.

Do NOT use NaCl. (Derusting is a separate bath from the Na_2SO_4 U-tube.)

Setup steps

1. Dissolve washing soda (or baking soda) in warm water; pour into a small jar (~100–250 mL).
2. Clip **rusty iron** nail/hardware to battery **cathode (–)**.
3. Clip **sacrificial steel strip or graphite** to battery **anode (+)**.
4. Immerse both; keep them from touching; confirm gentle bubbling.
5. Label group name; photograph the nail **before**.
6. Place a **control** rusty nail in plain water (no battery).
7. Leave powered overnight per instructor / venue rules.

End-of-class checklist

- [] All derusting cells connected and bubbling (or current confirmed)
- [] Control nails in plain water labeled
- [] Before photos taken
- [] Overnight power policy confirmed (prep room preferred)

Troubleshooting

Problem	Cause	Fix
No bubbles	Poor contact / weak electrolyte	Check spade connectors; remake ~0.1 M Na_2SO_4
One side only	Single electrode connected	Check both leads and polarity
Ratio far from 2:1	Uneven arms; leaks; late start	Equal fill levels; reseal stoppers; restart timer after steady bubbling
Chlorine smell	Salt used	Stop; dump; remake Na_2SO_4 bath
Pressure / liquid spray	Side arms sealed with no outlet	Open side arms or attach vent tubing
Graphite crumbling	High current / long run	Lower voltage; replace rods

Safety checklist

- ☐ No NaCl electrolyte in the U-tube
 - ☐ No flames near the apparatus (except instructor pop test)
 - ☐ Small gas volumes only
 - ☐ Pop test instructor-only, at a safe distance
 - ☐ Do not over-pressurize sealed limbs
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Experiment status

- ☐ Na₂SO₄ electrolyte recipe posted (~0.1 M amounts)
- ☐ U-tube apparatus pilot-tested (polarity, bubble rate, volume method)
- ☐ H₂ vs. O₂ side identified in pilot (cathode = H₂)
- ☐ Typical ratio from pilot: ____
- ☐ Rust-removal jars + washing soda staged for Part E

Syllabus Safety Notes

Apply these across all sessions. Session-specific hazards are listed in each session folder.

General rules

1. **No eating experimental materials.** Fruits, potatoes, and liquids used in experiments are no longer food.
2. **Avoid short circuits.** Batteries can heat up if directly shorted.
3. **Clean surfaces matter.** Electroplating requires clean metal surfaces.
4. **Metal salt waste should be collected.** Copper and silver solutions should not be casually poured down the drain.
5. **Water choice:** tap water is fine for most baths; use **distilled/deionized water for 0.01 M AgNO₃** (Session 3) so chloride does not precipitate silver.

Session-specific hazards

Session	Key hazard	Mitigation
1	Sharp nails/wires	Supervise handling; wash hands after metals
2	Copper solutions	Goggles; no ingestion; collect waste
3	Silver nitrate	Gloves + goggles; stains skin/clothing; small volumes
4	Gas evolution	No saltwater; no flames near setup; instructor-only pop test
5	Corrosion demos + electrolytic derusting	Use washing soda or baking soda only — NOT NaCl; disconnect power when not supervised

Hydrogen pop test (Session 4, instructor-only)

- Use a **tiny** volume of hydrogen only
- Perform away from the electrolysis setup
- **Never** ignite a sealed container or a mixed H₂/O₂ container